

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1 Objective

The objectives of this experiment are:

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet, and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2 Learning Outcomes

After completing this experiment students will be able to:

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet, and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3 Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4 Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogLeNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5 LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers, and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6 AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization, and GPU training.

Major Innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7 VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8 Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9 GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy et al. in 2014, introduced the Inception Module, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1 , 3×3 , 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10 ResNet

Increasing CNN depth often causes the vanishing gradient problem, making optimization difficult. ResNet solves this issue using skip (residual) connections, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

,

ResNet learns:

$$F(x) = H(x) - x$$

thus,

$$\text{Output} = F(x) + x$$

where:

- $F(x)$ = Residual Mapping
- x = Identity Mapping

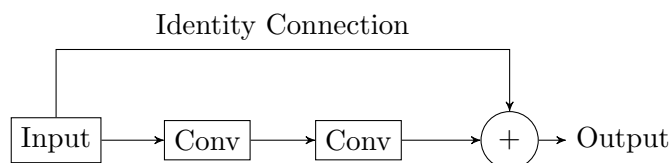


Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11 Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters. The dilation rate determines the spacing between kernel elements.

For a standard convolution:

$$D = 1$$

For dilated convolution:

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis
- Satellite imagery
- Object localization

12 Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13 Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch:

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

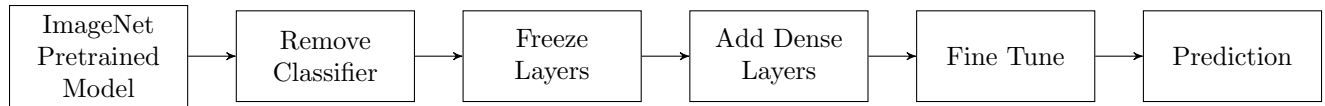


Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14 Dataset

Dataset: CIFAR-10

- Training Images: 50,000
- Testing Images: 10,000
- Number of Classes: 10
- Image Size: $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, and Truck.

15 Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range $[0, 1]$.
3. Display ten sample images.
4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models:

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps:

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using:

Hyperparameter	Value
Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine-tuning.

Task 5: Model Evaluation

Evaluate the trained model using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16 Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17 Source Code

Source: Github

Github Repository Link:

https://github.com/ssvibitha/DeepLearningLab_2026-27/blob/main/Lab4_TransferLearning/DL_Lab4.ipynb

18 Plots and Inference

1. Sample CIFAR-10 images

Sample CIFAR-10 Images



Figure 3: Sample Images

Inference:

The figures show sample images from the CIFAR-10 dataset. The images belong to 10 different categories such as Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck. Each image is 32 x 32 pixels and contains three color channels (RGB)

2. Training and Validation Accuracy

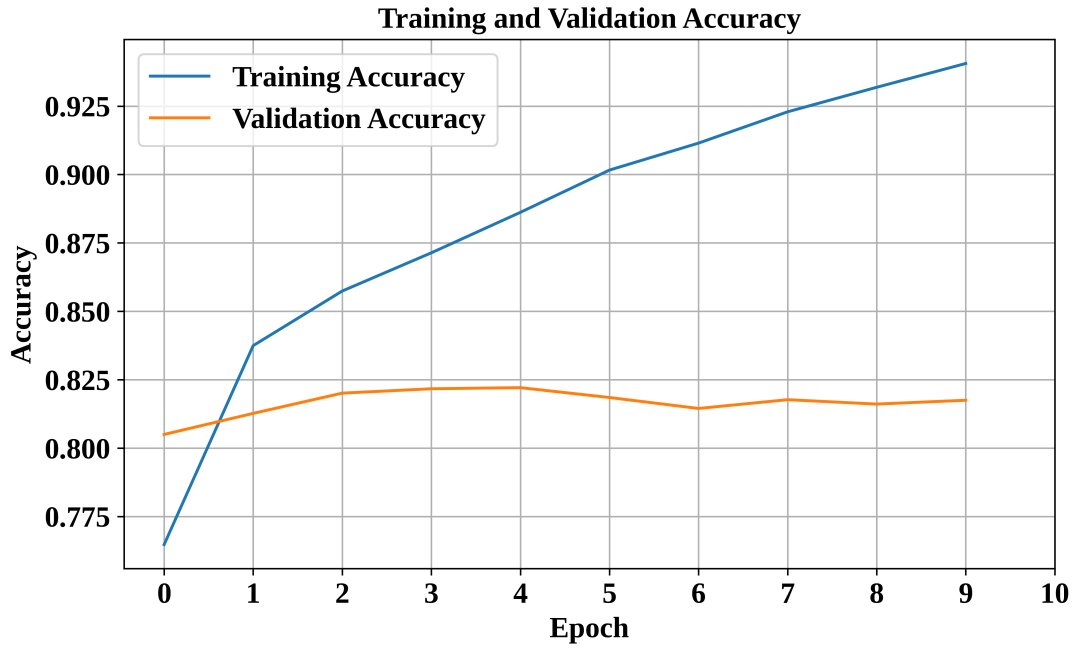


Figure 4: Training and Validation Accuracy

Inference:

Training accuracy increases smoothly from approximately 76 percent at Epoch 0 to over 92 percent by Epoch 9. This shows that the model effectively fits the training data with subsequent epochs.

Validation accuracy rises initially from around 80 percent and reaches its peak at approximately 82.5 percent around Epoch 3–4. After Epoch 4, the validation accuracy stabilizes.

After Epoch 4, training accuracy continues to rise while validation accuracy drops slightly and stabilizes. This shows that the model is overfitting.

3. Training and Validation Loss

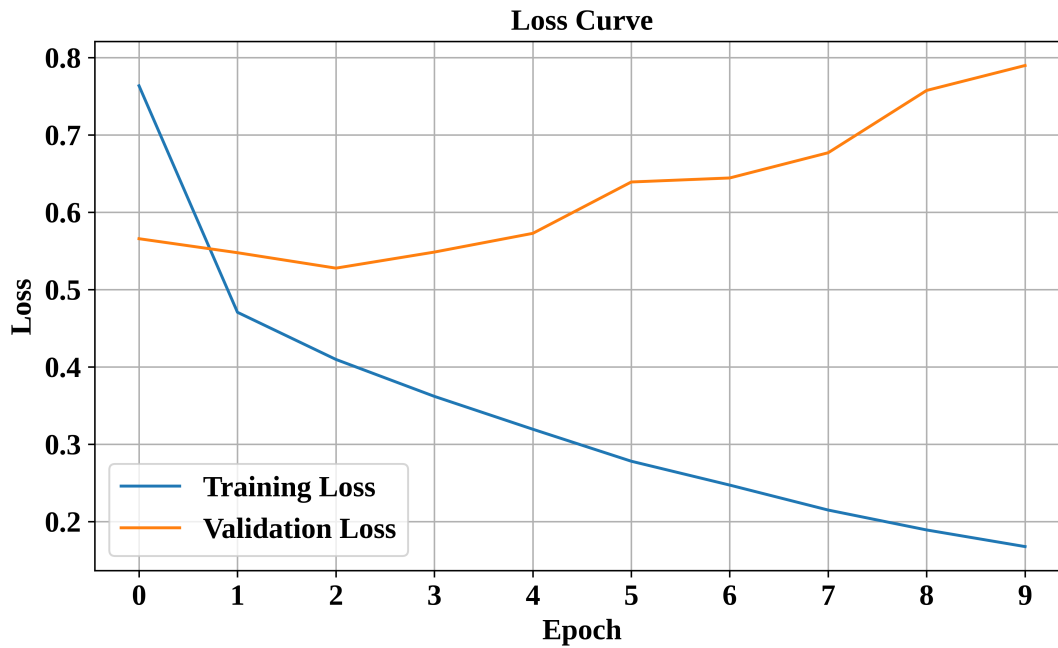


Figure 5: Training and Validation Loss

Inference:

The training loss sharply decreases from 75 percent to around 0.5 percent within the first 2 epochs. Then it steadily decreases for subsequent epochs.

Validation loss decreases initially reaching its lowest point with value about 50 percent around Epoch 2. Then the validation loss increases.

The gap between training and validation loss curves suggests that the model is learning the training data well but has limited generalization to unseen data.

4. Confusion Matrix

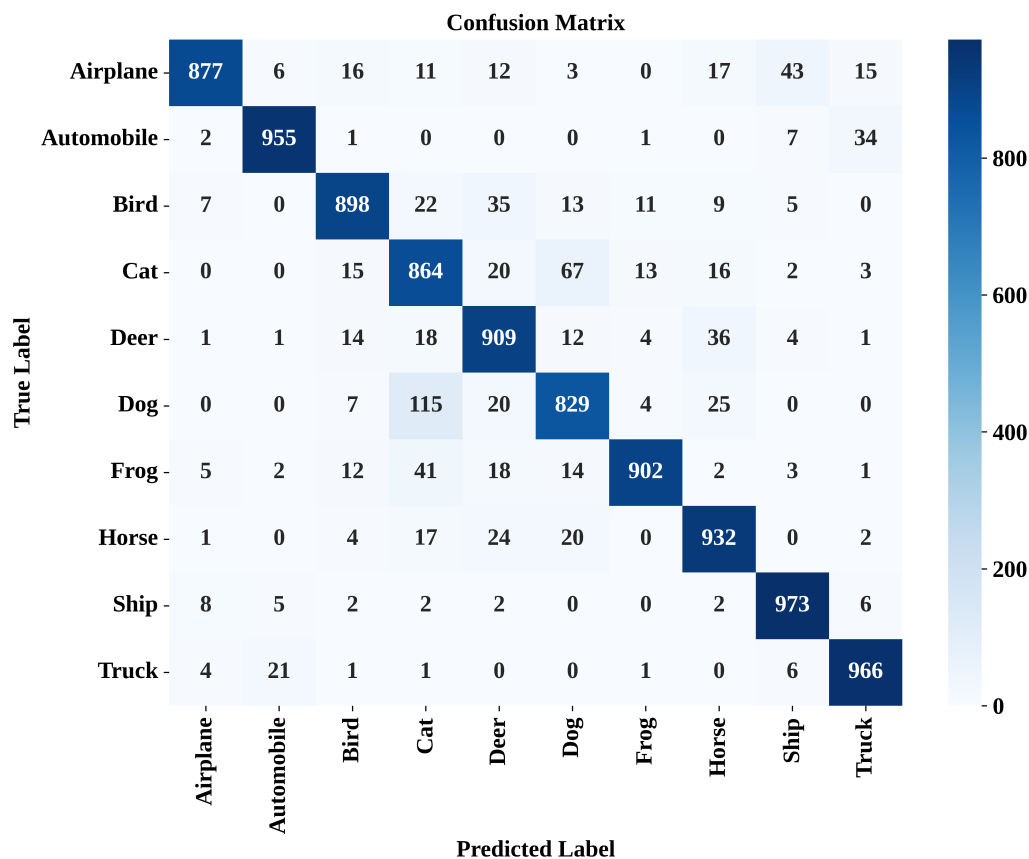


Figure 6: Confusion Matrix

Inference:

The confusion matrix shows that all the classes have been correctly classified most of the time. However 115 Dog images have been incorrectly classified as Cat. Also 67 Cat images have been incorrectly classified as Dog. In addition, 43 Airplane images have been misclassified as Ship. Also 34 images of Automobile has been misclassified as Truck. Moreover 36 Deer images has been misclassified as Horse.

This shows that the model has been able to predict most of the classes very well. However there is some misclassification among similar classes such as Dog and Cat, Deer and Horse, Automobile and Truck.

5. Misclassified Images

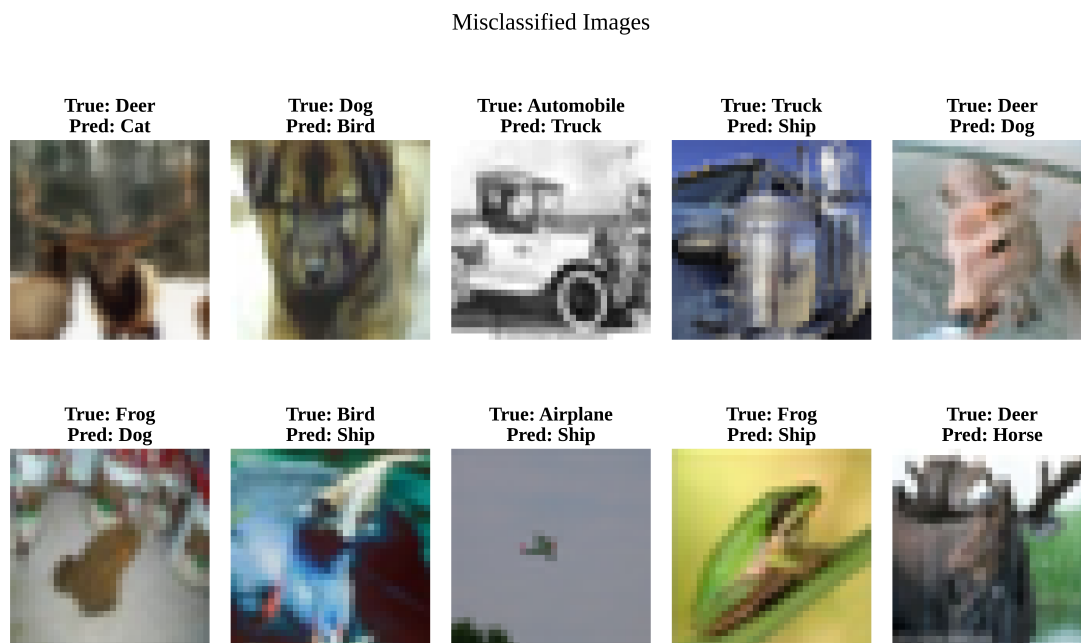


Figure 7: Misclassified Images

Inference:

The figure shows some of the images that has been misclassified by the model. From this figure we can see that Deer image has been misclassified as Dog and Horse. Similarly Frog image has been misclassified as Dog and Ship.

19 Results

19.1 Performance Metrics

Metric	Value
Training Accuracy	0.9406
Testing Accuracy	0.9105
Precision	0.9124063566237557
Recall	0.9105
F1-score	0.9107742104848716
Training Time	769.273529291153 seconds
Total Parameters	29,074,400

19.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
LeNet-5	83,126	54.45	77.5 seconds
AlexNet	36,925,322	71.19	343.1 seconds
VGG16	29,074,400	91.05	769.3 seconds
GoogleNet	22,066,346	69.58	296.1 seconds
ResNet50	23,851,274	86.38	392.7 seconds

19.3 Hyperparameter Study

VGG16

Learning Rate	Batch Size	Epochs	Optimizer	Dense Units	Frozen	Validation Accuracy	Time (sec)
0.001	32	10	Adam	128	All	0.8216	769.60
0.0001	32	10	Adam	128	All	0.8197	776.33
0.001	16	10	Adam	128	All	0.8195	772.22
0.001	64	10	Adam	128	All	0.8219	711.69
0.001	32	20	Adam	128	All	0.8248	1527.30
0.001	32	10	SGD	128	All	0.8018	767.58
0.001	32	10	Adam	256	All	0.8230	767.40
0.001	32	10	Adam	128	Partial	0.1000	1001.75

20 Discussion Questions

Answer the following questions:

1. Why is AlexNet considered a breakthrough in deep learning?

AlexNet was a major breakthrough because it demonstrated that a deep convolutional neural network (CNN) could achieve better performance on large-scale image classification. It used techniques like ReLU, dropout, data augmentation, and GPUs for faster training.

2. Why does VGG16 use only 3×3 convolution filters?

VGG16 uses 3×3 convolution filters because they require fewer parameters and computations than larger filters. Stacking two 3×3 filters gives an effective 5×5 receptive field, whereas stacking three 3×3 filters gives a 7×7 receptive field. This allows the network to capture larger patterns without using large filters directly. Each convolution is followed by ReLU, providing more nonlinearities and helping the network learn complex features. Using small filters also allows VGG16 to build a deeper network. Thus, 3×3 filters provide better feature learning with fewer parameters and greater depth.

3. Explain the advantages of the Inception module.

The Inception module processes an image using different operations like 1×1 , 3×3 , 5×5 convolutions and pooling in parallel. It can capture features at different scales, from small details to larger patterns. The outputs of these branches are combined to create a richer feature representation. Using 1×1 convolutions reduce the number of channels which lowers computational cost. Thus, Inception provides better feature extraction and computational efficiency.

4. What is the purpose of residual learning?

The main purpose of residual learning is to make very deep neural networks easier and more effective to train. Instead of learning the entire transformation, the network learns only the difference (residual) between the input and the desired output. Skip connections allow the original input to directly reach later layers, helping information and gradients flow smoothly. This helps prevent problems such as vanishing gradients and degradation, where adding more layers can actually reduce performance. It also makes the network easier to optimize because the layers only need to learn the necessary changes. Residual learning allows models like ResNet to become very deep while still maintaining good performance.

5. Differentiate LeNet and ResNet.

LeNet is a simple and shallow CNN mainly designed for handwritten digit recognition. On the other hand, ResNet is a much deeper CNN designed for complex image classification tasks. LeNet does not use skip connections, while ResNet can skip connections. LeNet has fewer parameters and requires less computation. Whereas, ResNet can have 18, 34, 50, 101, or 152 layers, making it much more powerful but computationally expensive.

6. What is Transfer Learning?

Transfer Learning is the process of using a model already trained on a large dataset for a new similar task. The pretrained model already knows useful features such as edges, textures, and shapes. These learned features are used instead of training a model from scratch. Here, the final classification layer is replaced according to the new task. It helps reduce training time, data requirements, and computational cost.

7. Why is fine tuning required?

Fine-tuning is used to adapt a pretrained model to a new task. A pretrained model already knows general features like edges, shapes, and textures, but it may not understand the specific features of the new task. Here we unfreeze some of the deeper layers and train them with the new data. A small learning rate is usually used so that the useful features already learned are not lost. This helps the model achieve better accuracy with less training time.

8. Explain the difference between dilated convolution and transpose convolution.

Dilated convolution increases the receptive field by adding gaps between kernel elements. It captures information from a larger area without greatly increasing the number of parameters. Transpose convolution is mainly used to increase the spatial size of feature maps. Dilated convolution is commonly used in segmentation, while transpose convolution is used in decoders and image generation.

9. Why do pretrained models converge faster?

Pretrained models start with already learned weights instead of random weights. They already know useful features such as edges, textures, shapes, and patterns. Therefore, the model does not need to learn these features from the beginning. It only needs to adapt the existing features to the new dataset or task. As a result, pretrained models generally require fewer epochs and less training time to achieve good performance.

10. Compare the computational complexity of LeNet and ResNet.

LeNet is a small and simple CNN with fewer layers and parameters so it requires less computation. ResNet is much deeper and has many more layers and parameters so it needs more computational power. As a result, ResNet requires more mathematical calculations, memory, and training time than LeNet. LeNet is lightweight and fast. Hence it can run easily on basic hardware. However ResNet is computationally expensive and works better with a GPU.

21 Additional Exercises

1. Implement transfer learning using VGG16.
2. Repeat the experiment using ResNet50.
3. Compare Adam and SGD optimizers.
4. Compare training with frozen layers and fine tuning.
5. Evaluate the model using Precision, Recall and F1-score.
6. Compare the performance of LeNet, AlexNet and ResNet.

22 References

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